Xiongyule Zhou

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Beijing Jiaotong University, No. 3
Shangyuancun, Haidian District, Beijing 100044,
China

EDUCATION

Beijing Jiaotong University, Beijing, China Sep 2023 – Jul 2027 (Expected)

Bachelor of Engineering in Mechatronics Engineering

Cumulative GPA: **3.67/4.0** (88.2/100)

Relevant Coursework: Artificial Intelligence, Robotics and Flexible Automation, Control Theory, Embedded Systems, Signals and Systems, Intelligent Mechatronic System Design

RESEARCH EXPERIENCE

Learning-Based Robotic Manipulation, Beijing Jiaotong University Sep 2025 – Feb 2026

- Developed a Python/PyTorch learning pipeline for a simulated robotic arm, using MATLAB/Simulink to model robot dynamics for grasping, target placement, and object-handoff tasks.
- Applied DINOv2 and OpenCV-based visual representations for vision-guided robotic manipulation, extracting object-level features for grasping, placement, and handoff tasks.
- Developed reinforcement learning pipelines for robotic manipulation, exploring policy optimization methods such as PPO/SAC for grasping and object-handoff tasks in simulated environments.
- Implemented imitation learning pipelines with demonstration-driven policies, exploring Diffusion Policy and behavior cloning methods for vision-guided robotic manipulation tasks.

Autonomous Robot Path Planning and Control, Beijing Jiaotong University Jun 2025 – Aug 2025

- Developed a host–embedded autonomous robot architecture, using a host computer for perception and navigation and an STM32 for real-time sensor acquisition, FSM logic, PID control, and motor execution.
- Tuned sensing thresholds and PID gains through repeated track testing, coordinating perception, planning, and control for autonomous route execution and obstacle avoidance.
- Prototyped an RGB-D 3D perception pipeline using Intel RealSense and Open3D, generating point clouds and estimating obstacle geometry for 3D-aware autonomous navigation.

Thermoelectric Autonomous Vehicle Design, Beijing Jiaotong University Nov 2024 – Apr 2025

- Designed a thermally powered autonomous vehicle using SolidWorks, developing cam and gear transmission mechanisms for thermal-to-mechanical energy conversion.
- Integrated line-following hardware and performed mechanical assembly, calibration, and system-level debugging to improve vehicle stability and reliability.

Reconfigurable Land–Air Amphibious UAV, Beijing Jiaotong University Mar 2024 – Aug 2024

- Developed and tuned the UAV flight-control system, focusing on attitude stabilization, takeoff/landing control, and PID parameter tuning through iterative whole-platform testing.
- Calibrated IMU and onboard sensors and processed three-axis acceleration and angular-velocity measurements, applying filtering and sensor-fusion methods to improve attitude estimation and control stability.
- Implemented and debugged ground–flight mode-switching logic, coordinating DC wheel motors, four independently actuated servos, and brushless propulsion motors for safe and responsive transitions between terrestrial driving and aerial flight.

WORK EXPERIENCE

Software Engineer Intern —Embodied AI, Anyverse Technology Co., Ltd., Aug 2026 – Present

- Developing Isaac Sim / Isaac Lab workflows for vision-based manipulation, synthetic robot-environment interaction, and embodied-learning experiments requiring high-fidelity simulation.
- Building a PyTorch–MuJoCo robot-learning pipeline integrating world models and VLA policies, connecting predictive dynamics, policy learning, and ROS 2 interfaces toward sim-to-real evaluation.
- Extended a C++/Python control SDK for a humanoid robot, adding encoder-state APIs, PVT trajectory execution, trajectory export, dynamics interfaces, and controller diagnostics for real-robot experimentation.
- Developing ROS 2 / MoveIt 2 manipulation pipelines for humanoid robots, integrating kinematic planning, trajectory optimization, and closed-loop execution in real-robot tasks.

Human Biomechanics & Embodied Simulation Intern , Chongqing Xingshijie Co., Ltd. Jan 2026 – Mar 2026

- Built human musculoskeletal models in OpenSim and imported human-motion datasets to establish a simulation workflow for analyzing whole-body movement and joint biomechanics.
- Computed and analyzed joint-level forces, loads, and kinematic profiles across human motions, identifying high-load phases and biomechanical patterns relevant to movement assessment and injury prevention.
- Translated simulation results into human-motion and biomechanical priors for embodied AI, providing joint-level references for human motion understanding, humanoid motion representation, and human-like robot control.

Robot Learning and Manipulation Intern, Csoft , Jul 2024 – Aug 2024

- Built a ROS/Ubuntu ALOHA bimanual robotic-arm system for teleoperation-based human-action imitation, supporting object grasping, transfer, book handling, and water-pouring tasks.
- Integrated dual-arm control, teleoperation, joint-state feedback, calibration, and demonstration execution into an end-to-end manipulation pipeline for reproducing human-operated behaviors.
- Diagnosed hardware–software synchronization and calibration errors through repeated real-robot trials, improving execution consistency for contact-rich grasping, transfer, and pouring behaviors.

AWARDS

- Second Prize, Beijing College Students Engineering Practice and Innovation Competition Mar 2025
- Third-Class Academic Excellence Scholarship 2024-2025, 2025-2026
- Third-Class Scholarship for Outstanding Social Work 2024-2025, 2025-2026
- Second-Class Sports Activity Scholarship 2023–2024

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Captain, Basketball Team, School of Mechanical Engineering Jun 2024 – Present

- Managed team operations, scheduling, and communication among players and coaching staff.
- Led the team to a top-four finish in the university basketball tournament.

Deputy Director, Visual Arts Department, University Arts Center Oct 2024 - Oct 2025

- Coordinated campus visual-media projects and produced photography/video content for university events.

SKILLS AND INTERESTS

Programming Languages: Python, C++, C, MATLAB
Robotics: ROS/ROS 2, Robot Manipulation, Motion Planning, Motion Control, Sim-to-Real
AI / ML: PyTorch, Deep Learning, Robot Learning, Computer Vision
Tools: Linux, Git, MATLAB/Simulink, MuJoCo / Isaac Sim
Languages: Chinese (Native), English (IELTS: 7.5)
Interests: Basketball (University Team Captain), Photography, Video Production, Reading, Robotics, Artificial Intelligence